

Dr. Zafar Ahmed, D.Sc.

Research publication portfolio

Affiliation	Bhabha Atomic Research Centre, Trombay, Mumbai, 400085
Research areas	One-dimensional quantum mechanics: Hermitian, non-Hermitian, PT-symmetric, and pseudo-Hermitian Hamiltonians
Scholar metrics	2160 citations; h-index 22; i10-index 38
Coverage	130 works listed on the public profile; dated records span 1986–2021
Profile	Google Scholar record

Year-wise publication matrix

Rows are shaded by complete year blocks. Position is Zafar Ahmed's place in the displayed abbreviated author list (position/total). Click an arXiv number to open the PDF.

Year	Author pos.	Authors (abbrev.)	Paper title	Received / preprint*	Published	Journal / publication	arXiv PDF
2021	1/2	Z Ahmed, PS Telkar	Families of curves orthogonal to the lines	2020-10-01*	–	arXiv preprint arXiv:2010.00939, Math. Gazette 105(563)306-309 (2021), 2021	2010.00939
2021	2/2	S Kumar, Z Ahmed	An update on coherent scattering from complex non-PT-symmetric Scarf II potential with new analytic forms	2019-12-30*	2021-04-16 (~1.3 yr*)	Pramana 95 (2), 1-6, 2021	1912.12876
2021	1/6	Z Ahmed, V Sharma, M Sharma, A Singhal, R Kaiwart, P Priyadarshini	Addendum to The paradoxical zero reflection at zero energy*(2017 Eur. J. Phys. 38 025401)	–	2021	European Journal of Physics 42 (4), 049401, 2021	–
2021	1/2	Z Ahmed, K Bhattacharya	Bound state eigenvalues from transmission coefficients	2021-03-12*	2021-07-29 (~4.6 mo*)	European Journal of Physics 42(5), 055405	2103.07096
2021	1/2	Z Ahmed, HF Jones	Scattering states and bound states of exponential potentials	2021-02-11*	–	arXiv preprint arXiv:2102.06095, 2021	2102.06095
2020	1/2	Z Ahmed, AN Joseph	An interesting track for the Brachistochrone	2020-10-29*	–	arXiv preprint arXiv:2010.15514, 2020	2010.15514
2020	1/3	Z Ahmed, S Kumar, D Sharma	Low reflection at zero or low-energies in the well-barrier scattering potentials	2020-05-07*	–	arXiv preprint arXiv:2005.03721, 2020	2005.03721
2019	1/4	Z Ahmed, S Kumar, D Ghosh, JA Nathan	Addendum to Divergence of in discontinuous potential wells'(2018 Eur. J. Phys. 39 055402)	–	2019	European Journal of Physics 41 (1), 019401, 2019	–
2019	1/4	Z Ahmed, S Kumar, D Ghosh, T Goswami	Solvable model of bound states in the continuum (BIC) in one dimension	2019-01-31*	2019-08-07 (~6.2 mo*)	Physica Scripta 94 (10), 105214, 2019	1901.11340
2019	1/4	Z Ahmed, S Kumar, T Goswami, S Hajirnis	Symmetric Fermi-type potential	2019-04-03*	–	arXiv preprint arXiv:1904.02284, 2019	1904.02284
2018	1/3	Z Ahmed, S Kumar, D Ghosh	Three types of discrete energy eigenvalues in complex -symmetric scattering potentials	2018-06-18*	2018	Physical Review A 98 (4), 042101, 2018	1806.06578
2018	1/2	Z Ahmed, S Kumar	PT-symmetric potentials with imaginary asymptotic saturation	2018-08-17*	2022-07-26 (~3.9 yr*)	Pramana 96(3)	1808.05828
2018	1/4	Z Ahmed, S Kumar, D Ghosh, JA Nathan	Divergence of in discontinuous potential wells	–	2018	European Journal of Physics 39 (5), 055402, 2018	–
2018	1/2	Z Ahmed, S Kumar	Pearson's correlation coefficient in the theory of networks: A comment	2018-03-16*	–	arXiv preprint arXiv:1803.06937, 2018	1803.06937
2018	1/3	Z Ahmed, D Ghosh, S Kumar	Coherent scattering from semi-infinite non-Hermitian potentials	2017-09-25*	2018-02-16 (~4.7 mo*)	Physical Review A 97 (2), 023828, 2018	1709.08513
2018	2/2	S Kumar, Z Ahmed	Pseudo-symmetric random matrices	2018-02-13*	–	arXiv preprint arXiv:1802.04588, 2018	1802.04588
2018	1/4	Z Ahmed, D Ghosh, S Kumar, N Turumella	Solvable models of an open well and a bottomless barrier: one-dimensional exponential potentials	–	2018-01-29	European Journal of Physics 39 (2), 025404, 2018	–
2018	1/4	Z Ahmed, D Ghosh, S Kumar, JA Nathan	Expectation values of and in the square well potential	2018-01-15*	–	arXiv preprint arXiv:1801.04730, 2018	1801.04730
2018	1/4	Z Ahmed, S Kumar, A Kumar, M Irfan	Coalescing versus merging of energy levels in one-dimensional potentials	2017-02-27*	2018-05-24 (~1.2 yr*)	XXII DAE High Energy Physics Symposium, 345-348, 2018	1702.08335

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2017	2/2	S Kumar, Z Ahmed	Pseudosymmetric random matrices: Semi-Poisson and sub-Wigner statistics	2017-05-25*	2017-08-30 (~3.2 mo*)	Physical Review E 96 (2), 022157, 2017	1705.09179
2017	1/3	Z Ahmed, S Kumar, D Sharma	Dirichlet spectra of the paradigm model of complex PT-symmetric potential: $V(x) = -i(x)N$	2016-06-15*	2017 (~6.6 mo*)	Annals of Physics 383, 635-644, 2017	1606.04757
2017	1/6	Z Ahmed, V Sharma, M Sharma, A Singhal, R Kaiwart, P Priyadarshini	Reply to Comment on The paradoxical zero reflection at zero energy ¹	–	2017	European Journal of Physics 38 (3), 038003, 2017	–
2017	1/1	Z Ahmed	Orthogonal Functions	–	2017	AMERICAN MATHEMATICAL MONTHLY 124 (5), 466-467, 2017	–
2017	2/2	S Kumar, Z Ahmed	Spectral statistics for ensembles of various real random matrices	2017-04-10*	–	arXiv preprint arXiv:1704.02715, 2017	1704.02715
2017	2/2	D Ghosh, Z Ahmed	Scattering from a semi-infinite non-Hermitian potential	2017-09-25*	2017	Proceedings of the DAE Symp. on Nucl. Phys 62, 458, 2017	1709.08513
2016	1/6	Z Ahmed, V Sharma, M Sharma, A Singhal, R Kaiwart, P Priyadarshini	The paradoxical zero reflection at zero energy	2016-05-24*	2016	European Journal of Physics 38 (2), 025401, 2016	1605.07315
2016	1/4	Z Ahmed, D Sharma, R Kaiwart, M Irfan	Supersymmetric partner potentials arising from nodeless half bound states	2016-12-21*	–	arXiv preprint arXiv:1612.07081, 2016	1612.07081
2016	1/4	Z Ahmed, M Irfan, A Kumar, A Singhal	Hermitian Hamiltonians: Matrix versus Schrödinger's	2016-07-13*	–	arXiv preprint arXiv:1608.01543, 2016	1608.01543
2016	1/4	Z Ahmed, S Kumar, M Sharma, V Sharma	Revisiting double Dirac delta potential	2016-03-23*	2016-06-10 (~2.6 mo*)	European Journal of Physics 37 (4), 045406, 2016	1603.07726
2016	1/1	Z Ahmed	Bound state eigenfunctions need to vanish faster than	2016-05-19*	2016-05-27 (~8 d*)	European Journal of Physics 37 (4), 045404, 2016	1605.05902
2016	1/3	Z Ahmed, JA Nathan, D Ghosh	Transparency of the complex PT-symmetric potentials for coherent injection	2014-10-21*	2016-02 (~1.3 yr*)	Physics Letters A 380 (4), 562-566, 2016	1410.5530
2016	1/4	Z Ahmed, JA Nathan, D Sharma, D Ghosh	Real discrete spectrum of complex PT-symmetric scattering potentials	2015-10-21*	2016-05-28 (~7.2 mo*)	Non-Hermitian Hamiltonians in Quantum Physics, 1-9, 2016	1510.06226
2016	1/4	Z Ahmed, JA Nathan, D Sharma, D Ghosh	Vol. 184. Non-Hermitian Hamiltonians in Quantum Physics-Selected Contributions from the 15th International Conference on Non-Hermitian Hamiltonians in Quantum Physics, 2015 ...	–	2016	Publication details unavailable on profile	–
2016	1/3	Z Ahmed, JA Nathan, D Ghosh	Transparency of the PT-symmetric complex potentials for coherent injection	2014-10-21*	–	arXiv preprint arXiv:1410.5530, Phys. Lett. A 380 (2016) 562, 2016	1410.5530
2015	1/4	Z Ahmed, D Ghosh, JA Nathan, G Parkar	Accidental crossings of eigenvalues in the one-dimensional complex PT-symmetric Scarf-II potential	2015-03-09*	2015-10 (~6.8 mo*)	Physics Letters A 379 (39), 2424-2429, 2015	1503.02426
2015	1/3	Z Ahmed, D Ghosh, JA Nathan	A new solvable complex PT-symmetric potential	2015-02-17*	2015-08 (~5.4 mo*)	Physics Letters A 379 (28-29), 1639-1642, 2015	1502.04838
2015	1/4	Z Ahmed, S Pavaskar, D Sharma, L Prakash	Avoided crossings of energy levels in one-dimension	2015-08-04*	–	arXiv preprint arXiv:1508.00661, 2015	1508.00661
2015	1/3	Z Ahmed, S Pavaskar, L Prakash	New shape resonances in one dimension	2015-04-01*	2015-05-20 (~1.6 mo*)	European Journal of Physics 36 (4), 048001, 2015	1504.00115
2015	1/2	Z Ahmed, JA Nathan	Real discrete spectrum in the complex non-PT-symmetric Scarf II potential	2014-11-12*	2015-05 (~5.6 mo*)	Physics Letters A 379 (10-11), 865-869, 2015	1411.3231

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2014	2/3	SS Pany, Z Ahmed, BP Dubey	Analysis of magnetic electron lens with secant hyperbolic field distribution	2014-05-19*	2014	Journal of Instrumentation 9 (12), P12009, 2014	1405.4653
2014	1/1	Z Ahmed	Ahmed's Integral: the maiden solution	2014-11-19*	—	arXiv preprint arXiv:1411.5169, 2014	1411.5169
2014	1/1	Z Ahmed	Coherent perfect absorption with and without lasing in complex potentials	2014-04-07*	2014-09-10 (~ 5.1 mo*)	Journal of Physics A: Mathematical and Theoretical 47 (38), 385303, 2014	1404.1679
2014	1/3	Z Ahmed, L Prakash, S Pavaskar	Non-catastrophic resonant states in one dimensional scattering from a rising exponential potential	2014-08-11*	—	arXiv preprint arXiv:1408.2367, 2014	1408.2367
2014	1/3	Z Ahmed, S Pavaskar, L Prakash	One dimensional scattering from two-piece rising potentials: a new avenue of resonances	2014-08-01*	—	arXiv preprint arXiv:1408.0231, 2014	1408.0231
2014	1/4	Z Ahmed, S Pavaskar, I Yadav, T Bhatia	Anti-symmetric square well and barrier potential between two rigid walls	2014-06-18*	—	arXiv preprint arXiv:1406.4761, 2014	1406.4761
2014	1/2	Z Ahmed, I Yadav	Position-momentum uncertainty products	2014-03-12*	2014-05-09 (~ 1.9 mo*)	European Journal of Physics 35(4), 045015	1403.2824
2014	1/4	Z Ahmed, T Bhatia, S Pavaskar, A Kumar	Special rectangular (double-well and hole) potentials	2014-01-31*	—	arXiv preprint arXiv:1401.8070, 2014	1401.8070
2013	1/2	Z Ahmed, S Kesari	The simplest model of the zero-curvature eigenstate	—	2013-12-13	European Journal of Physics 35 (1), 018002, 2013	—
2013	1/1	Z Ahmed	Non-unitarity and non-reciprocity in scattering from real potentials in presence of confined non-linearity	2013-12-11*	—	arXiv preprint arXiv:1312.3157, 2013	1312.3157
2013	1/1	Z Ahmed	PT-symmetry and Transparency	2013-08-06*	—	arXiv preprint arXiv:1308.1270, 2013	1308.1270
2013	1/1	Z Ahmed	Reciprocity and unitarity in scattering from a non-Hermitian complex PT-symmetric potential	2012-07-30*	2013-05 (~ 9.0 mo*)	Physics Letters A 377 (13), 957-959, 2013	1207.6896
2013	1/2	Z Ahmed, AN Savita	Lines of force of the electric point dipole	—	2013-02-12	European Journal of Physics 34 (2), L39, 2013	—
2012	4/4	A Ghatak, JA Nathan, BP Mandal, Z Ahmed	Spectral singularity and reflectionlessness in non-Hermitian (complex) Ginocchio potential	—	2012-10-31	Journal of Physics A: Mathematical and Theoretical 45 (46), 465305, 2012	—
2012	3/3	A Ghatak, BP Mandal, Z Ahmed	Spectral singularity and deep multiple minima in the reflectivity in non-Hermitian (complex) Ginocchio potential	2012-07-09*	—	arXiv preprint arXiv:1207.1979, 2012	1207.1979
2011	1/1	Z Ahmed	New features of scattering from a one-dimensional non-Hermitian (complex) potential	2011-10-20*	2011-12-19 (~ 2.0 mo*)	Journal of Physics A: Mathematical and Theoretical 45 (3), 032004, 2011	1110.4485
2011	1/1	Z Ahmed	Transmission through a non-overlapping well adjacent to a finite barrier	2011-09-27*	—	arXiv preprint arXiv:1109.5792, 2011	1109.5792
2011	1/1	Z Ahmed	Comment on "Resonant states and transmission coefficient oscillations for potential wells and barriers," by A. Uma Maheswari, P. Prema, and CS Shastry [Am. J. Phys. 78 (4), 412 ...	—	2011-05-13	American Journal of Physics 79 (6), 682-683, 2011	—
2010	1/1	Z Ahmed	Studying the scattering length by varying the depth of the potential well	—	2010-03-11	American Journal of Physics 78 (4), 418-421, 2010	—

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2009	1/1	Z Ahmed	Zero width resonance (spectral singularity) in a complex PT-symmetric potential	2009-08-20*	2009-11-09 (~2.7 mo*)	Journal of Physics A: Mathematical and Theoretical 42 (47), 472005, 2009	0908.2876
2009	1/1	Z Ahmed	Scarcity of real discrete eigenvalues in non-analytic complex-symmetric potentials	—	2009	Pramana 73 (2), 323-328, 2009	—
2009	1/1	Z Ahmed	Scarcity of real discrete eigenvalues in non-analytic complex InlineEquation ID=" IEq1"> ImageObject FileRef=" 120432009124ArticleIEq1. gif" Format=" GIF" Color=" BlackWhit ...	—	2009	Pramana 73 (2), 323, 2009	—
2007	1/1	Z Ahmed	A note on Coulomb scattering amplitude	2003-10-03*	2007-07-20 (~3.8 yr*)	International Journal of Modern Physics A 22 (18), 3131-3136, 2007	quant-ph/0310019
2007	1/1	Z Ahmed	Eigenvalue problems for the complex PT-symmetric potential $V(x) = igx$	2006-09-28*	2007-04 (~6.1 mo*)	Physics Letters A 364 (1), 12-16, 2007	quant-ph/0609219
2006	1/1	Z Ahmed	An ansatz for the eigenstates in PT-symmetric quantum mechanics	2006-08-10*	2006-12 (~3.7 mo*)	Physics Letters A 360 (2), 238-242, 2006	quant-ph/0608091
2006	1/1	Z Ahmed	An interesting model of quantal transmission	—	2006-09-10	International Journal of Modern Physics A 21 (22), 4439-4447, 2006	—
2006	1/1	Z Ahmed	PT-symmetry in conventional quantum physics	—	2006-07-26	Journal of Physics A: Mathematical and General 39 (32), 9965, 2006	—
2006	1/1	Z Ahmed	Reflectionlessness, kurtosis and top curvature of potential barriers	—	2006-05-23	Journal of Physics A: Mathematical and General 39 (23), 7341, 2006	—
2006	1/2	Z Ahmed, SR Jain	A pseudo-unitary ensemble of random matrices, PT-symmetry and the Riemann hypothesis	2004-07-21*	2006-02-10 (~1.6 yr*)	Modern Physics Letters A 21 (04), 331-338, 2006	quant-ph/0407167
2005	1/1	Z Ahmed	Novel phase-space orbits and quantization	—	2005-10-12	Journal of Physics A: Mathematical and General 38 (43), L701, 2005	—
2005	1/3	Z Ahmed, CM Bender, MV Berry	Reflectionless potentials and symmetry	2005-08-16*	2005-09-14 (~29 d*)	Journal of Physics A: Mathematical and General 38 (39), L627, 2005	quant-ph/0508117
2005	1/1	Z Ahmed	Phase space trajectories and quantization in PT-symmetric quantum mechanics	—	2005-09	Czechoslovak Journal of Physics 55 (9), 1055-1062, 2005	—
2005	1/2	Z Ahmed, MA Prasad	An application of the Gaussian integral	—	2005-02-01	AMERICAN MATHEMATICAL MONTHLY 112 (2), 182-183, 2005	—
2005	2/2	G Lamb, Z Ahmed	An Application of the Gaussian Integral: 11011	—	2005-02-01	The American Mathematical Monthly 112 (2), 182-183, 2005	—
2004	4/5	G Lamb, C Stallmann, G Thompson, Z Ahmed, OP Lossers	An Integral Three Ways: 11000	—	2004-12	The American Mathematical Monthly 111 (10), 918-920, 2004	—
2004	1/1	Z Ahmed	Gaussian-random ensembles of pseudo-Hermitian matrices	2004-07-20*	2004-10 (~2.4 mo*)	Czechoslovak journal of physics 54 (10), 1011-1018, 2004	quant-ph/0407154
2004	1/1	Z Ahmed	Handedness of complex PT-symmetric potential barriers	2003-12-18*	2004-04 (~3.4 mo*)	Physics Letters A 324 (2-3), 152-158, 2004	quant-ph/0312151
2004	1/2	Z Ahmed, SR Jain	Number of quantal resonances	2003-09-29*	2004-01-06 (~3.3 mo*)	Journal of Physics A: Mathematical and General 37 (3), 867, 2004	quant-ph/0309206

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2003	1/1	Z Ahmed	Pseudo-reality and pseudo-adjointness of Hamiltonians	2003-06-13*	2003-10-01 (~3.6 mo*)	Journal of Physics A: Mathematical and General 36 (41), 10325, 2003	quant-ph/0306093
2003	1/1	Z Ahmed	C-, PT-and CPT-invariance of pseudo-Hermitian Hamiltonians	2003-02-19*	2003	Journal of Physics A: Mathematical and General 36 (37), 9711, 2003	quant-ph/0302141
2003	1/2	Z Ahmed, SR Jain	Pseudounitary symmetry and the Gaussian pseudounitary ensemble of random matrices	2002-09-30*	2003-04-30 (~7.0 mo*)	Physical Review E 67 (4), 045106, 2003	quant-ph/0209165
2003	1/1	Z Ahmed	P-, T-, PT-, and CPT-invariance of hermitian hamiltonians	2003-02-11*	2003-04 (~1.6 mo*)	Physics Letters A 310 (2-3), 139-142, 2003	quant-ph/0302084
2003	1/2	Z Ahmed, SR Jain	Gaussian ensemble of 2x 2 pseudo-Hermitian random matrices	–	2003-03-13	Journal of Physics A: Mathematical and General 36 (12), 3349, 2003	–
2003	1/1	Z Ahmed	An ensemble of non-Hermitian Gaussian-random 2x 2 matrices admitting the Wigner surmise	–	2003-02	Physics Letters A 308 (2-3), 140-142, 2003	–
2003	1/2	Z Ahmed, SR Jain	PART 2-RAPID COMMUNICATIONS-General methods of statistical physics-Pseudounitary symmetry and the Gaussian pseudounitary ensemble of random matrices	–	2003	Physical Review-Section E-Statistical Nonlinear and Soft Matter Physics 67 ..., 2003	–
2002	1/3	Z Ahmed, K Dale, GL Lamb	Definitely an Integral: 10884	–	2002-08	The American Mathematical Monthly 109 (7), 670-671, 2002	–
2002	1/1	Z Ahmed	Pseudo-Hermiticity of Hamiltonians under gauge-like transformation: real spectrum of non-Hermitian Hamiltonians	–	2002-03	Physics Letters A 294 (5-6), 287-291, 2002	–
2002	1/1	Z Ahmed	Left and right handedness of Schroedinger reflection from a -symmetric complex potential	–	2002	DAE-BRNS symposium on nuclear physics: contributed papers. V. 45B, 2002	–
2002	1/1	Z Ahmed	Discrete symmetries, pseudo-hermiticity and pseudo-unitarity	–	2002	DAE-BRNS symposium on nuclear physics: invited talks. V. 45A, 2002	–
2002	2/3	SR Jain, Z Ahmed, RG Thomas	On the non-Lorentzian shape of hadron resonances	–	2002	DAE-BRNS symposium on nuclear physics: contributed papers. V. 45B, 2002	–
2001	1/1	Z Ahmed	Pseudo-Hermiticity of Hamiltonians under imaginary shift of the coordinate: real spectrum of complex potentials	2001-08-03*	2001-11 (~3.0 mo*)	Physics Letters A 290 (1-2), 19-22, 2001	quant-ph/0108016
2001	1/1	Z Ahmed	Schrodinger transmission through one-dimensional complex potentials	–	2001-09	Physical Review A 64 (4), 042716, 2001	–
2001	1/1	Z Ahmed	Energy band structure due to a complex, periodic, PT-invariant potential	–	2001-07	Physics Letters A 286 (4), 231-235, 2001	–
2001	3/4	KA Kearnes, A Szenderei, Z Ahmed, MA Prasad	A difference of square roots	–	2001	AMERICAN MATHEMATICAL MONTHLY 108 (6), 570-570, 2001	–
2001	1/1	Z Ahmed	10884	–	2001-06-01	The American Mathematical Monthly 108 (6), 566-566, 2001	–
2001	4/?	HAS Ali, KA Kearnes, A Szenderei, Z Ahmed, MA Prasad, ...	A Difference of Square Roots: 10806	–	2001	American Mathematical Monthly, 570-570, 2001	–
2001	1/1	Z Ahmed	Real and complex discrete eigenvalues in an exactly solvable one-dimensional complex PT-invariant potential	–	2001-04	Physics Letters A 282 (6), 343-348, 2001	–
2001	1/1	Z Ahmed	Quasi-bound state in supersymmetric quantum mechanics	–	2001-03	Physics Letters A 281 (4), 213-217, 2001	–
2001	1/1	Z Ahmed	Addendum to:"Real and complex discrete eigenvalues in an exactly solvable one-dimensional complex PT-invariant potential"-[Phys. Lett. A 282 (2001) 343]	–	2001	Physics Letters A 3 (287), 295-296, 2001	–
2000	5/5	M Hajja, P Walker, D Arrigo, X Wang, Z Ahmed	A Cubic Spherical Integral: 10771	–	2000-12	The American Mathematical Monthly 107 (10), 954-955, 2000	–
2000	5/5	SS Malik, M Dudeja, Amita, AK JAIN, Z Ahmed	Semiclassical Quantisation of Particle-Rotor Model and I= 2 Staggering of Superdeformed Bands	–	2000-12	International Journal of Modern Physics E 9 (06), 487-506, 2000	–
2000	3/3	D Arrigo, XS Wang, Z Ahmed	A cubic spherical integral	–	2000	AMERICAN MATHEMATICAL MONTHLY 107 (10), 954-955, 2000	–
2000	1/2	Z Ahmed, PM Jarvis	A family of integrals: 10777	–	2000-12	The American Mathematical Monthly 107 (10), 956-957, 2000	–
2000	1/1	Z Ahmed	Reflection from an interface	–	2000-04-28	Journal of Physics A: Mathematical and General 33 (16), 3161, 2000	–
2000	2/2	SR Jain, Z Ahmed	Brownian motion model of random matrices revisited	1998-04-23*	2000-04 (~1.9 yr*)	Physics Letters A 268 (1-2), 25-30, 2000	cond-mat/9804253
2000	1/2	Z Ahmed, SR Jain	Random matrix model for disordered conductors	–	2000-03	Pramana 54 (3), 413-422, 2000	–
2000	4/?	R Helmbold, W Kim, RA Agnew, Z Ahmed, S Amghibech, KF Andersen, ...	The Variance of Logarithms: 10690	–	2000	American Mathematical Monthly, 182-182, 2000	–
2000	1/1	Z Ahmed	Resonances in supersymmetric quantum mechanics	–	2000	International symposium on nuclear physics. V. 43B, 2000	–
2000	1/1	Z Ahmed	10777	–	2000-01-01	The American Mathematical Monthly 107 (1), 83-83, 2000	–
2000	1/1	Z AHMED	SEMICLASSICAL QUANTISATION OF PARTICLE-ROTOR MODEL AND I= 2 STAGGERING OF SUPERDEFORMED BANDS	–	2000-12	International Journal of Modern Physics E 9 (6), 487-506, 2000	–
1999	1/1	Z Ahmed	Reflection above potential steps	–	1999-01-01	Journal of Physics A: Mathematical and General 32 (15), 2767, 1999	–
1999	1/1	Z Ahmed	Asymmetric parabolic barrier as a method in heavy-ion fusion	–	1999	DAE symposium on nuclear physics: contributed papers. V. 42B, 1999	–

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1998	1/1	Z Ahmed	Tunnelling through asymmetric parabolic potential barriers	–	1997-07-21	Journal of Physics A: Mathematical and General 31 (13), 3115, 1998	–
1998	1/1	Z Ahmed	Reflection from a composite potential well	–	1997-12	Physics Letters A 6 (239), 411, 1998	–
1998	4/4	M Dudeja, AK Jain, SS Malik, Z Ahmed	I= 2 staggering in superdeformed bands and nonlinear features in the particle-rotor model	–	1998	DAE symposium on nuclear physics. V. 41B: contributed papers, 1998	–
1997	1/1	Z Ahmed	Reflection from a composite potential well	–	1997-12	Physics Letters A 236 (4), 289-295, 1997	–
1997	1/1	Z Ahmed	Tunnelling through asymmetric parabolic potential barriers	–	1997-07-21	Journal of Physics A: Mathematical, Nuclear and General 30 (14), 5243-5244, 1997	–
1997	4/4	AK Jain, M Dudeja, SS Malik, Z Ahmed	Nonlinear dynamics of particle rotor model and superdeformed bands	–	1997-01	Physics Letters B 392 (3-4), 243-248, 1997	–
1996	1/1	Z Ahmed	* Tunnelling through Assymmetric Parabolic Potential Barriers*, de Zafar Ahmed (Bhabha Atomic Research Centre, Bombay)	–	1996	Publication details unavailable on profile	–
1996	3/3	SR Jain, AK Jain, Z Ahmed	Nonlinear dynamics of high-j cranking model: A semi-classical approach	–	1996-03	Physics Letters B 370 (1-2), 1-4, 1996	–
1996	4/5	AK Jain, M Dudeja, SR Jain, Z Ahmed, SS Malik	Superdeformed bands as the rigid-rotor bands	–	1996	DAE symposium on nuclear physics: contributed papers. Volume 39B (1996), 1996	–
1996	1/1	Z Ahmed	Unusual reflection from step barriers	–	1996-01	Physics Letters A 210 (1-2), 1-4, 1996	–
1996	3/4	M Dudeja, AK Jain, Z Ahmed, SS Malik	First evidence of a nonlinear effect in nuclear spectra	–	1996	DAE symposium on nuclear physics: contributed papers. Volume 39B (1996), 1996	–
1995	2/2	BK Nayak, Z Ahmed	Commenton On: 'Intermittancy in Quark-Gluon-Plasma Phase Transitions'	–	1995-09-18	Phys. Rev. Lett. 75 (12), 2248, 1995	–
1994	1/2	Z Ahmed, BJ Roy	Barrier tunnelling in neutron transfer reactions	–	1994	DAE symposium on nuclear physics. Vol. 37B (1994): Contributed papers, 1994	–
1993	1/1	Z Ahmed	Tunneling through a one-dimensional potential barrier	–	1993-06-01	Physical Review A 47 (6), 4761, 1993	–
1991	1/1	Z Ahmed	Tunnelling through the Morse barrier	–	1991-07	Physics Letters A 157 (1), 1-5, 1991	–
1986	1/2	Z Ahmed, RC Sethi	AN ANALYTICAL METHOD to DETERMINE the GEOMETRY of the SECTORS	–	1986	Proceedings of the Eleventh International Conference on Cyclotrons and their ..., 1986	–
n.d.	1/1	Z Ahmed	A solvable potential and a new penetrability formula for nucleus-nucleus fusion	–	–	Publication details unavailable on profile	–
n.d.	1/1	Z Ahmed	A simple two-channel barrier penetration model of heavy ion fusion	–	–	Publication details unavailable on profile	–
n.d.	1/1	Z Ahmed	A variable-height parabolic barrier for heavy ion fusion	–	–	Publication details unavailable on profile	–

Reading notes. Alternating pale-blue and white bands apply to complete profile-year groups. A starred first date is the first arXiv submission date; its starred bracket is a preprint-to-publication interval, not publisher turnaround. Publisher received dates are shown only when independently available; otherwise a dash is retained. Durations use days only below 30 days, months below one year, and years from one year onward. A dash means that the corresponding date or arXiv record was not identified. “n/a” means that the displayed author string does not identify Zafar Ahmed. Profile years are retained even where publisher metadata gives a different online-publication year.

Data note. The public profile contains duplicate preprint/journal records, variant titles, incomplete entries, and at least one apparently mismatched authorship. Inclusion here reports the displayed profile and is not an independent authorship verification.

Data snapshot. Public Google Scholar profile, Crossref publication metadata, and arXiv records, checked 21 July 2026. Scholar metrics are dynamic and may change after this snapshot.